

Real-time auctions for digital advertising poles

GeoBid lets advertisers find high-footfall digital poles on a map and bid for 2-hour advertising slots in a live auction. The operator (GeoBid admin) runs the poles and auctions and sees the business in a finance dashboard.

65 poles

on 15 major roads around Kondapur, Hyderabad

12 slots a day

2-hour slots, open for bidding 7 days ahead

4 seats per slot

a rolling ad shared by 4 advertisers

2 rounds

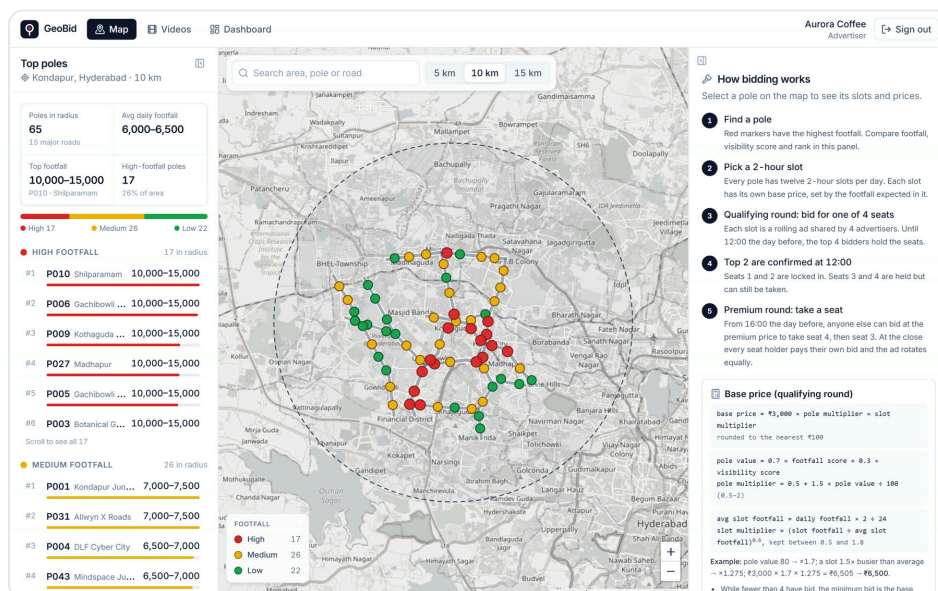
qualifying, then a premium round for the last seats

Live updates

bids appear instantly for everyone watching

129 tests

automated backend tests, all passing



The map an advertiser sees: poles coloured by footfall (red = high, amber = medium, green = low), top poles per category on the left, and the bidding guide on the right.

What GeoBid does

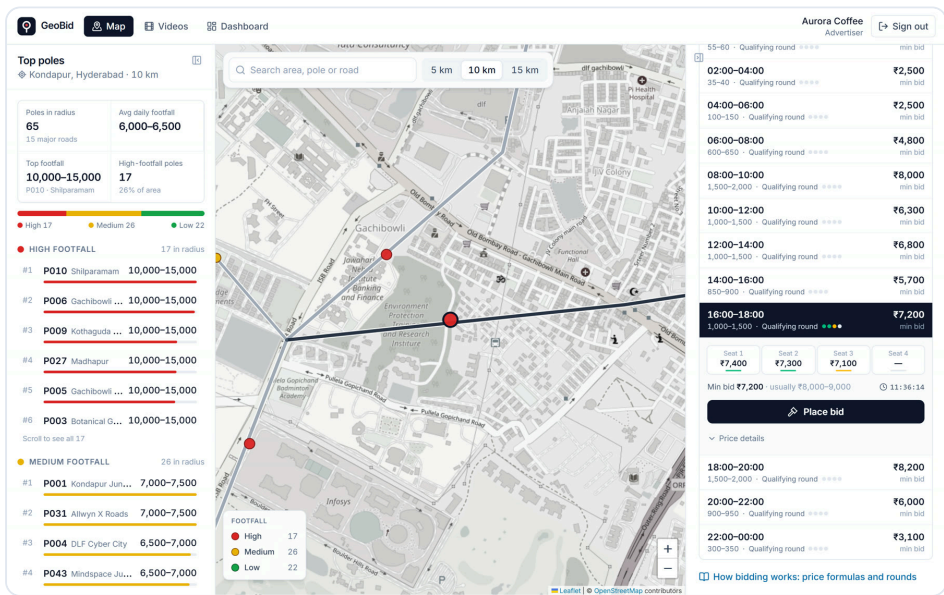
Digital poles on busy roads can show several advertisers' ads in rotation. Pricing a slot well is hard: the same pole is worth far more during the evening rush than at 3 a.m. GeoBid prices every 2-hour slot from the footfall expected during that slot and then lets advertisers compete for it in a transparent, time-boxed auction.

Who uses it

Role	What they do
Advertiser	Explores poles on the map, compares footfall and prices, bids for seats, and follows their seats, bid history and wins in a personal dashboard. New advertisers can sign up themselves.
GeoBid admin	Runs the platform: sees each pole's status and performance on the map, opens, advances and closes auctions, and follows revenue in the Finance dashboard. Admins don't bid.

What's in the proof of concept

- **Interactive map** of 65 poles with footfall categories, a 5 / 10 / 15 km radius around any location, search by area, pole or road, and ranked pole lists.
- **Per-slot pricing**: every 2-hour slot has its own base price from that slot's expected footfall, plus a 60-day price history and an expected price range.
- **Two-round seat auction** with live updates over WebSockets and an automatic worker that moves rounds on schedule.
- **Advertiser dashboard**: seats held, seats won (with footfall during each slot and that day), bid history, and an inventory finder with filters.
- **Admin tools**: pole analysis and auction controls on the map, and a finance dashboard (daily / weekly / monthly revenue, top advertisers and poles, frequent buyers).
- **Footfall videos**: sample camera footage with detection overlays, viewable in the app.



Selecting a pole shows its 12 slots for the chosen day, each with expected footfall and the current minimum bid. The selected 16:00–18:00 slot shows its 4 seats and the Place bid button.

How a slot is sold

Each 2-hour slot is a rolling ad shared by **4 advertisers** (seats). Ads rotate equally and every seat holder pays **their own winning bid**. Bidding opens 7 days ahead and runs in two rounds on the day before the ad date:

Qualifying round	12:00, day before	Break	Premium round	Close
Opens up to 7 days ahead Anyone bids from the base price. The top 4 bidders hold the seats.	Top 2 confirmed Seats 1–2 are locked in. Seats 3–4 are held but can still be taken.	12:00 – 16:00 No bidding.	16:00 until 2 h before the slot Anyone else can bid at the premium price to take seat 4, then seat 3.	2 hours before the slot Seat holders win and pay their own bids.

Bidding rules

- While fewer than 4 have bid, the minimum bid is the base price.
- Once all 4 seats are held, the minimum is the lowest seat bid + ₹100.
- Raising your own bid: your bid + ₹100.
- **No matching bids:** a bid can't be within ₹100 of another seat holder's bid.
- The 2 confirmed bidders can't bid in the premium round.
- A new premium bid takes seat 4, then seat 3; the bumped bidder can bid again.
- Rounds move on automatically; the admin can also advance or close an auction.

Place a bid

P014 ● HIGH ● Live 1

Tomorrow · Wed, 7 Oct · 16:00–18:00 · 1,000–1,500 footfall

Qualifying round ⌚ Qualifying ends in 11:36:12

Seat 1	Bidder 5	₹7,400
Seat 2	Bidder 3	₹7,300
Seat 3	Bidder 4	₹7,100
Seat 4	Empty	Open

4 ads rotate equally; each seat pays its own bid. Base ₹7,100 · usually ₹8,000–9,000 on a Wed.

₹ 7200 Place bid

Min ₹7,200 +₹500 +₹1,000

Recent bids

Bidder 5	₹7,400
Bidder 3	₹7,300
Bidder 4	₹7,100

The bid popup: live seats, round timer, the expected price range for that weekday and slot, and the minimum bid.

Base and premium prices

The **base price** is where bidding starts in the qualifying round. It combines how valuable the pole is with how busy that particular 2-hour slot is:

```
base price = ₹3,000 × pole multiplier × slot multiplier (rounded to the nearest ₹100)
pole value = 0.7 × footfall score + 0.3 × visibility score (0-100)
pole multiplier = 0.5 + 1.5 × pole value ÷ 100 (0.5-2.0)
average slot footfall = daily footfall × 2 ÷ 24
slot multiplier = (slot footfall ÷ average slot footfall)0.6 (kept between 0.5 and 1.8)
```

Example: a pole with value 80 gets a ×1.7 pole multiplier. A slot 1.5× busier than the pole's average slot gets ×1.275. So ₹3,000 × 1.7 × 1.275 = ₹6,503, and the base price is **₹6,500**.

The **premium price** opens the premium round:

```
premium price = 1.5 × top qualifying bid (rounded up to the next ₹100)
```

Example: with a top qualifying bid of ₹7,700, the premium price is 1.5 × ₹7,700 = ₹11,550, rounded up to **₹11,600**. If seat 3 or 4 is empty or still held by a qualifying bid, that's the minimum; once both are held by premium bids, the minimum is the lower premium bid + ₹100.

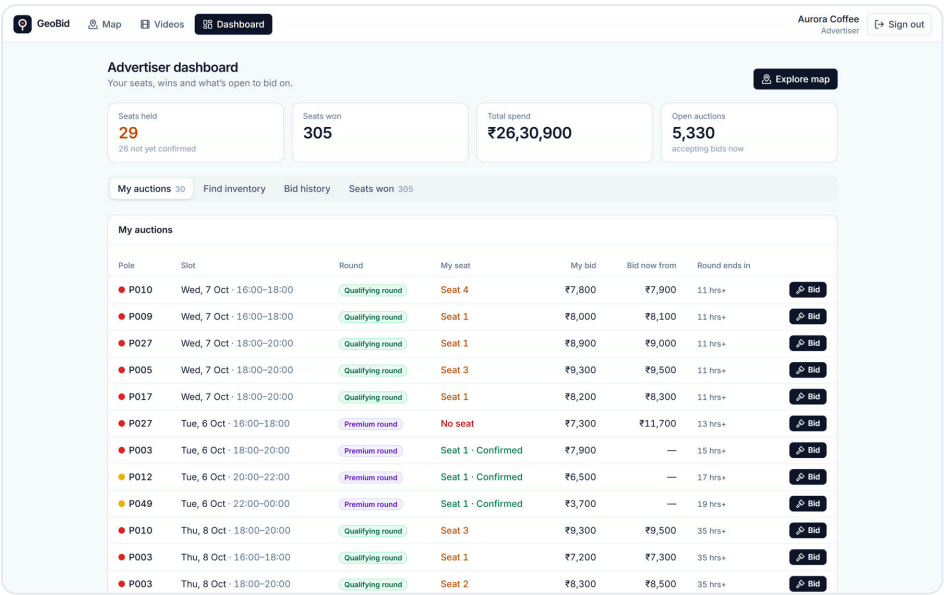
Footfall shown to users

Footfall is shown as a range (for example 1,000–1,500 rather than 1,287) per 2-hour slot, by weekday. Poles are grouped into high, medium and low footfall relative to the other poles in the area. The pole panel also shows a 60-day price history for the slot and the usual price range for that weekday.

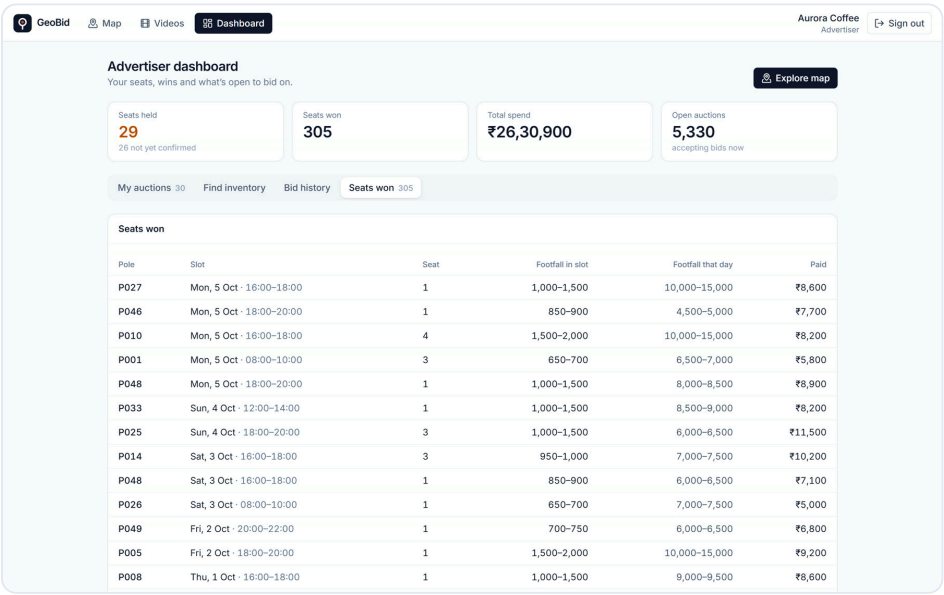
About the data. Footfall figures in this proof of concept are synthetic, produced by a generator behind a replaceable "footfall provider" interface, so real counts (for example from cameras) can be plugged in later without changing the rest of the system. Prices and the formula are a POC business rule, not an industry standard.

Finding, bidding and tracking

- **Map and lists:** poles by footfall category; the left panel lists every pole in the radius per category (scrollable), the right panel shows the selected pole or the bidding guide with the exact formulas.
- **Bidding in a popup** from the slot list, with live seats and timers; the guide is one click away from any pole.
- **Dashboard tabs:** My auctions (seat, my bid, the bid needed now, time left), Find inventory (filters by date, slot, footfall, score, price, round, radius), Bid history (every bid and its result), Seats won.
- **Seats won with footfall:** each win shows the footfall during that slot and over that whole day.
- **Sign-up:** new advertisers create an account and start bidding straight away.



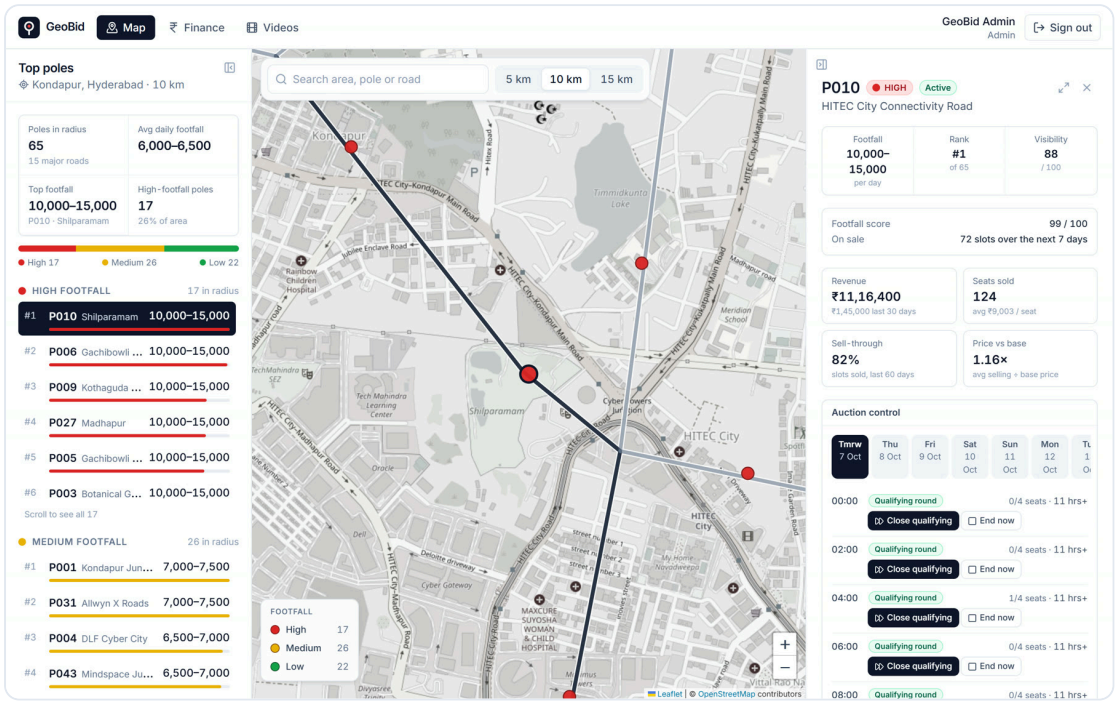
Advertiser dashboard for a demo advertiser (Aurora Coffee): seats held, seats won, total spend and open auctions, then tabs for auctions, inventory, bid history and wins.



Seats won, with the footfall during the slot and over the whole day.

Running poles and auctions

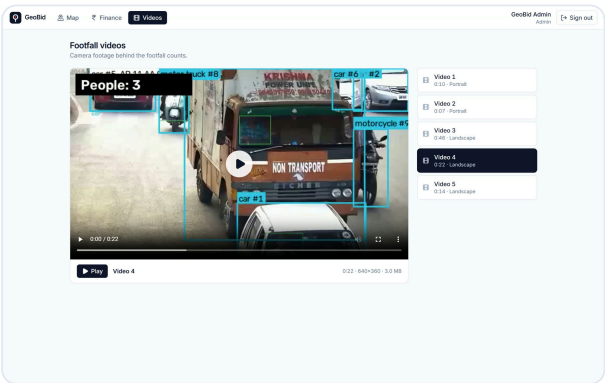
For the admin, clicking a pole opens its **analysis and auction controls** instead of the bidding panel: status, footfall, rank and visibility; revenue, seats sold, sell-through and price versus base; then **Auction control** for every slot over the next 7 days, where the admin can open a scheduled auction, move it to the next round (close qualifying → open premium → close), end it immediately, or create one for a slot without an auction. Every action needs a second click to confirm.



Admin view of pole P010: status and performance, then auction control per slot with round, seats filled and time until the round moves on.

Footfall videos

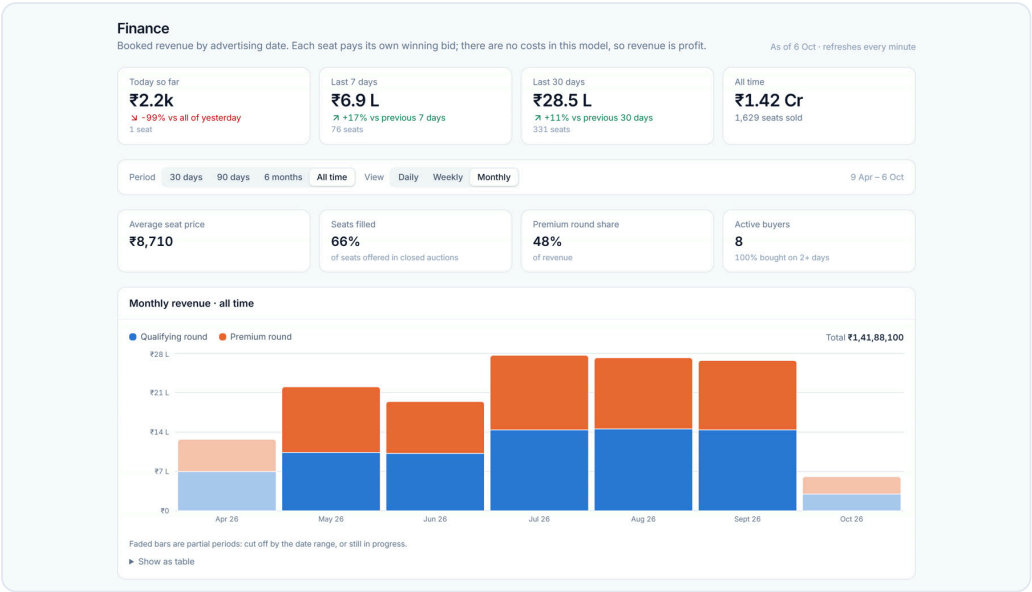
The Videos page plays sample camera footage with detection overlays (vehicles and people counted), illustrating how footfall could be measured. The videos are converted to a browser-friendly format and stream with seeking. They are not yet connected to the footfall numbers in the app.



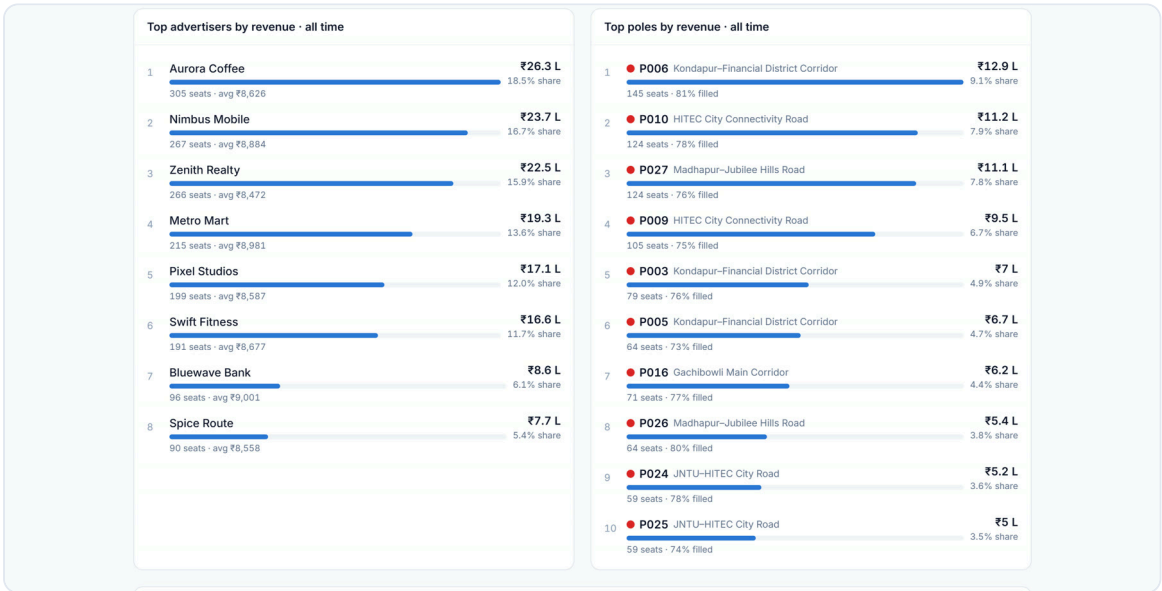
The Videos page.

Revenue at a glance

The admin's Finance page shows revenue today so far, over the last 7 and 30 days (each against the period before) and all time. One filter row sets the period (30 days, 90 days, 6 months or all time) and the view (daily, weekly or monthly) for everything below: revenue over time split into qualifying- and premium-round seats, average seat price, seats filled, premium share, active buyers, top advertisers and poles, frequent buyers, and revenue by time of day and footfall category. Revenue is booked on the advertising date; there are no costs in this model, so revenue equals profit.



Headline revenue and monthly revenue split by round. Faded bars are partial periods (the first month is cut off by the data range; the current month is in progress). The screenshot was taken just after midnight, so "Today so far" is small.



Top advertisers and top poles by revenue, with share, seats, average price and how full each pole's seats were.

What the demo shows

The demo is rebuilt from scratch on every start and follows the real clock, so auctions are always in a realistic state. It contains 6 months of completed sales and an auction for every slot of every pole for the next 7 days. All purchases come from the **8 demo advertisers**; the admin's Finance totals equal the sum of their dashboards' spend (this is checked by an automated test).

Figure (all time, at the time of writing)	Value
Revenue	₹1.42 Cr
Seats sold	1,629
Revenue in the last 30 days	₹28.5 L (331 seats)
Average seat price	₹8,710
Share of revenue from premium-round seats	48%
Share of revenue from high-footfall poles	76%
Best-earning slots	18:00–20:00, then 16:00–18:00
Top advertiser	Aurora Coffee, ₹26.3 L (305 seats)

Figures change as the clock moves and auctions close; they come from synthetic data and are for demonstration only.

Demo accounts

Account	Role	Notes
admin@geobid.local	Admin	Pole analysis, auction control, Finance
advertiser1@geobid.local ... advertiser8@geobid.local	Advertiser	6 months of purchases, about 30 live auctions each
newuser@geobid.local	Advertiser	No bids or purchases, for testing a first-time bidder

Password for all demo accounts: geobid123. New advertisers can also sign up at </signup>.

How it's built

Part	Technology
Backend	Python, FastAPI, SQLAlchemy, SQLite; JWT sign-in; WebSockets for live auction updates; a background worker that moves auction rounds on schedule; a caching map-tile proxy.
Frontend	React, TypeScript, Vite, Tailwind CSS, React Router; Leaflet for the map; Recharts for charts.
Data	Synthetic footfall per pole, weekday and hour behind a footfall-provider interface; generated price history and demo auctions.
Hosting	Frontend on Vercel, backend on Render, both deployed from GitHub.
Quality	129 automated backend tests (auctions and seat rules, pricing, footfall, dashboards and finance, live updates, videos, sign-up), all passing.

How the pieces fit

- The browser talks to the backend over a REST API for data and bids, and over a WebSocket per auction for instant updates (with automatic reconnection and polling as a fallback).
- Bids are validated against the seat rules on the server; concurrent bids are handled safely, so two advertisers can't take the same seat at the same price.
- A background worker checks every second for auctions whose round should move on, and closes auctions on time.

Limitations and next steps

Today (proof of concept)	Next step
Footfall is synthetic	Connect real counts (for example from the camera detection shown on the Videos page) through the footfall-provider interface.
No payments or invoices	Add payment collection for won seats and invoices.
No ad creative upload	Let winners upload and approve the ad they want to show.
No cost model	Add pole operating costs to show margin, not just revenue.
SQLite, free hosting	Move to a managed database (for example PostgreSQL) and paid hosting; the free backend sleeps when idle and takes about a minute to wake.
Videos are public	Restrict footage to signed-in users; it shows number plates and people.
Demo data resets on restart	Keep a persistent database once real data is used.